

# Yuzhou Joey Zou

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## Research Interests

Microlocal Analysis, Inverse Problems, Partial Differential Equations, Mathematical Physics.

## Employment

2025–        Oakland University, Assistant Professor  
2022–2025   Northwestern University, Boas Assistant Professor  
              *Postdoctoral mentor:* Jared Wunsch  
2021–2022   University of California, Santa Cruz, Postdoctoral Scholar-Employee  
              *Postdoctoral mentor:* François Monard

## Education

2016–2021   Stanford University, Ph.D. in Mathematics  
              *Thesis advisor:* András Vasy  
              *Thesis title:* *Microlocal analysis with applications to seismic inverse problems*  
2012–2016   University of Chicago, B.S. with honors in Mathematics and B.A. in Chemistry  
              *GPA:* 3.83

## Publications

- [13] “**Gaussian-weighted normal operators on Euclidean space**”. Preprint, 2026.  
arXiv:2605.02232
- [12] “**A Comparison Test for Meromorphic Extensions**”, joint work with Adi Glücksam. Preprint, 2026. arXiv:2601.05896
- [11] “**Tensor tomography on asymptotically hyperbolic surfaces**”, joint work with Nikolas Eptaminitakis and François Monard. Preprint, 2025. arXiv:2510.04144.
- [10] “**A Gutzwiller trace formula for singular potentials**”, joint work with Jared Wunsch and Mengxuan Yang. Preprint, 2025. arXiv:2509.05220.
- [9] “**Asymptotic Expansion of the Eigenvalues of a Bathtub Potential with Quadratic Ends**”. *Communications in Mathematical Physics*, Vol. 406, no. 220. 2025. DOI: 10.1007/s00220-025-05394-3
- [8] “**The hyperbolic X-ray transform: new range characterizations, mapping properties and functional relations**”, joint work with Nikolas Eptaminitakis and François Monard. *Inverse Problems and Imaging*, Vol. 22, 65-100. 2026. DOI:10.3934/ipi.2025045

- [7] “**Helmholtz quasi-resonances are unstable under most single-signed perturbations of the wave speed**”, joint work with Euan A. Spence and Jared Wunsch. *Journal of Differential Equations*, Vol. 440, Part 2. 2025. DOI:10.1016/j.jde.2025.113441.
- [6] “**The Morse index theorem for mechanical systems with reflections**”, joint work with Jared Wunsch and Mengxuan Yang. *Nonlinearity*, Vol. 37, no. 8. 2024. DOI: 10.1088/1361-6544/ad5636.
- [5] “**Boundary triples for a family of degenerate elliptic operators of Keldysh type**”, joint work with François Monard. *Pure and Applied Analysis*, Vol. 6, no. 2, 541-580. 2024. DOI: 10.2140/paa.2024.6.541.
- [4] “**The  $C^\infty$ -isomorphism property for a class of singularly-weighted X-ray transforms**”, joint work with Rohit K. Mishra and François Monard. *Inverse Problems*, Vol. 39, no. 2. 2023. DOI: 10.1088/1361-6420/aca8cb.
- [3] “**Microlocal Methods for The Elastic Travel Time Tomography Problem for Transversely Isotropic Media**”. Preprint, 2021. arXiv:2112.14455.
- [2] “**Streak artifacts from non-convex metal objects in X-ray tomography**”, joint work with Yiran Wang. *Pure and Applied Analysis*, Vol. 3, no. 2, 295-318. 2021. DOI: 10.2140/paa.2021.3.295.
- [1] “**Partial Global Recovery in the Elastic Travel Time Tomography Problem for Transversely Isotropic Media**”. *Annales de l’Institut Fourier*, Vol. 74, no. 5, 2077-2139. 2024. DOI: 10.5802/aif.3617.

## Expository Papers

- [2] “**Entropy and kinetic formulations of conservation laws**”.  
Written at the University of Chicago Mathematics REU 2015.
- [1] “**Modes of convergence for Fourier series**”.  
Written at the University of Chicago Mathematics REU 2014.

## Awards and Honors

|            |                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------|
| 2026       | Faculty Research Fellowship, Oakland University                                                                          |
| 2021       | Mathematics Distinguished Service Award, Dept. of Mathematics, Stanford University                                       |
| 2019       | Robert Osserman Teaching Award, Dept. of Mathematics, Stanford University                                                |
| 2018, 2016 | Honorable Mention, NSF Graduate Research Fellowship                                                                      |
| 2016       | Paul R. Cohen Memorial Prize, University of Chicago Dept. of Mathematics<br>Awarded to top graduating mathematics majors |
| 2016       | 1st prize at the 23rd International Mathematics Competition, Blagoevgrad, Bulgaria<br>(19th place overall)               |
| 2015       | Honorable Mention, Putnam Exam                                                                                           |

## Teaching

### Instructor, Oakland University

*Duties: write and give lectures, write homework and exams.*

|        |      |          |             |
|--------|------|----------|-------------|
| Summer | 2026 | MTH 1441 | Precalculus |
| Winter | 2026 | MTH 1554 | Calculus 1  |
| Fall   | 2025 | MTH 1554 | Calculus 1  |

## **Instructor, Northwestern University**

*Duties: write and give lectures, write homework and exams.*

|         |      |            |                                                     |
|---------|------|------------|-----------------------------------------------------|
| Spring* | 2025 | Math 228-1 | Multivariable Differential Calculus for Engineering |
| Winter  | 2025 | Math 220-2 | Single-Variable Integral Calculus                   |
| Spring* | 2024 | Math 220-2 | Single-Variable Integral Calculus                   |
| Winter  | 2024 | Math 220-2 | Single-Variable Integral Calculus                   |
| Winter  | 2023 | Math 230-2 | Multivariable Integral Calculus                     |
| Fall    | 2022 | Math 220-1 | Single-Variable Differential Calculus               |

\* – co-lead instructor

## **Instructor, University of California, Santa Cruz**

*Duties: write and give lectures, write homework and exams.*

|        |      |           |                                                                  |
|--------|------|-----------|------------------------------------------------------------------|
| Spring | 2022 | Math 218  | Advanced Parabolic and Hyperbolic Partial Differential Equations |
| Winter | 2022 | Math 121A | Differential Geometry                                            |

## **Instructor, Stanford University**

*Duties: write and give lectures, write homework and exams.*

|                     |      |         |                            |
|---------------------|------|---------|----------------------------|
| Summer <sup>◊</sup> | 2021 | Math 19 | Single Variable Calculus 1 |
| Summer <sup>◊</sup> | 2020 | Math 19 | Single Variable Calculus 1 |

<sup>◊</sup> – conducted online

## **Administrative Teaching Assistant, Stanford University**

*Duties: manage course logistics for a large (~ 300 students) course (e.g. arrange exam logistics, manage homework/exam grading, maintain course website, answer student emails, etc.), hold office hours, grade exams.*

|                     |      |         |                                           |
|---------------------|------|---------|-------------------------------------------|
| Spring <sup>◊</sup> | 2021 | Math 51 | Linear Algebra and Multivariable Calculus |
| Winter              | 2020 | Math 51 | Linear Algebra and Multivariable Calculus |
| Autumn              | 2018 | Math 51 | Linear Algebra and Multivariable Calculus |

<sup>◊</sup> – conducted online

## **Teaching Assistant, Stanford University**

*Duties: lead discussion sections, hold office hours, grade exams.*

|        |      |         |                                           |
|--------|------|---------|-------------------------------------------|
| Winter | 2018 | Math 51 | Linear Algebra and Multivariable Calculus |
|--------|------|---------|-------------------------------------------|

## **Course Assistant, Stanford University**

*Duties: hold office hours, grade homework and exams, write solutions.*

|        |      |           |                                  |
|--------|------|-----------|----------------------------------|
| Autumn | 2019 | Math 205A | Graduate Real Analysis 1         |
| Summer | 2019 | Math 19   | Single Variable Calculus 1       |
| Winter | 2019 | Math 205B | Graduate Real Analysis 2         |
| Autumn | 2017 | Math 171  | Fundamental Concepts of Analysis |

|        |      |          |                                           |
|--------|------|----------|-------------------------------------------|
| Spring | 2017 | Math 172 | Lebesgue Integration and Fourier Analysis |
| Autumn | 2016 | Math 20  | Single Variable Calculus 2                |

## Graduate Assistant, Stanford Online High School

*Duties: help various aspects of Stanford Online High School operations, including researching high school math curricula, grading for various courses, etc..*

|        |      |                                 |
|--------|------|---------------------------------|
| Winter | 2021 | Research and Teaching Assistant |
| Autumn | 2020 | Research and Teaching Assistant |

## Reader, University of Chicago

*Duties: grade homework.*

|        |      |          |                              |
|--------|------|----------|------------------------------|
| Winter | 2016 | Math 255 | Abstract Algebra 2           |
| Autumn | 2015 | Math 254 | Abstract Algebra 1           |
| Spring | 2015 | Math 205 | Analysis in $\mathbb{R}^n$ 3 |
| Winter | 2015 | Math 204 | Analysis in $\mathbb{R}^n$ 2 |
| Autumn | 2014 | Math 203 | Analysis in $\mathbb{R}^n$ 1 |

## Junior Tutor, University of Chicago

*Duties: lead discussion section, grade homework.*

|        |      |          |                                     |
|--------|------|----------|-------------------------------------|
| Spring | 2014 | Math 133 | Elementary Functions and Calculus 3 |
| Winter | 2014 | Math 132 | Elementary Functions and Calculus 2 |
| Autumn | 2013 | Math 131 | Elementary Functions and Calculus 1 |

## Conferences Organized

|     |      |                                                                                                                                  |  |
|-----|------|----------------------------------------------------------------------------------------------------------------------------------|--|
| Mar | 2026 | Special Session on Microlocal Analysis and Mathematical Physics<br>2026 AMS Spring Eastern Sectional Meeting, Boston College, MA |  |
| Jun | 2024 | Microlocal Analysis and Quantum Dynamics<br>Summer school and conference, Northwestern University                                |  |

## Seminars Organized

|             |                                                          |  |  |
|-------------|----------------------------------------------------------|--|--|
| 2023-2024   | Analysis Seminar, Northwestern University (co-organized) |  |  |
| Winter 2018 | Student Analysis Seminar, Stanford University            |  |  |
| Autumn 2017 | Kiddie Colloquium, Stanford University                   |  |  |

## Invited Conference Talks

|     |      |                                                                                                                                       |  |
|-----|------|---------------------------------------------------------------------------------------------------------------------------------------|--|
| May | 2026 | May Midwestern Microlocal Meeting, Chicago                                                                                            |  |
| Jan | 2026 | Special Session on <i>Recent Trend of Mathematical Modeling and Methods in Medical Applications</i> , Joint Mathematics Meetings 2026 |  |
| Jun | 2025 | Great Lakes Mathematical Physics Meeting, University of Kentucky                                                                      |  |

Mar 2025 Special Session on *Inverse Problems: Theory and Applications*, 2025 AMS Spring Central Sectional  
 Dec 2024 Harmonic and Microlocal Analysis in Partial Differential Equations, MATRIX  
 Oct 2024 Special Session on *Harmonic Analysis, Partial Differential Equations, and Spectral Theory*, 2024 AMS Fall Western Sectional  
 Oct 2024 Triangle Area Inverse Problems Weekend, NC State University  
 Aug 2024 Geometric Inverse Problems Summer School, UC Santa Cruz  
 Jun 2024 Great Lakes Mathematical Physics Meeting, Michigan State University  
 May 2024 SIAM Conference on Imaging Science, Atlanta  
 Mar 2024 Ohio River Analysis Meeting, University of Kentucky  
 Feb 2024 Texas Analysis and Mathematical Physics Symposium, Texas A&M University  
 Oct 2023 Spectral Theory and Applications, Texas A&M University  
 Sep 2023 Applied Inverse Problems 2023, Göttingen, Germany  
 Jun 2023 Special Session on *Inverse Problems and Imaging*, The 13th AIMS Conference on Dynamical Systems, Differential Equations and Applications  
 Aug 2022 Inverse Problems in Analysis and Geometry, Helsinki  
 Jul 2022 Workshop on Microlocal Analysis & PDEs, University College London  
 Dec 2021 Session on *Geometric Tomography and Microlocal Analysis*, 2021 CMS Winter Meeting  
 Aug 2021 Inverse problems and nonlinearity, Helsinki

## Invited Seminar Talks

May 2025 Analysis Seminar, Northwestern University  
 Mar 2025 Colloquium, Oakland University  
 Oct 2024 Baby Inverse Problems Seminar (online)  
 Oct 2024 Inverse Problems Seminar, UC Irvine  
 Sep 2024 Analysis and Applied Mathematics Seminar, University of Illinois, Chicago  
 Dec 2023 Spectral and Scattering Theory Seminar, Purdue University  
 Nov 2023 University College London  
 Oct 2023 Analysis & PDE Seminar, UC Berkeley  
 Oct 2023 Geometry & Analysis Seminar, UC Santa Cruz  
 Oct 2023 Analysis & PDE Seminar, Stanford University  
 Sep 2023 Analysis & Differential Geometry Seminar, Emory University  
 Apr 2023 PDE Seminar, Northwestern University  
 Apr 2023 Inverse Problems Seminar, University of Washington  
 Oct 2022 Analysis Seminar, Northwestern University  
 Oct 2022 Geometry and Topology Seminar, NC State University  
 May 2022 Analysis and PDE Seminar, University of Kentucky  
 Apr 2022 International Zoom Inverse Problems Seminar, UC Irvine  
 Oct 2021 Geometry and Analysis Seminar, UC Santa Cruz  
 Sep 2021 HADES Seminar, UC Berkeley  
 May 2020 Geometry and Analysis Seminar, UC Santa Cruz  
 Mar 2020 Differential Geometry & PDE Seminar, University of Washington  
 Feb 2020 Analysis & PDE Seminar, Stanford University

Feb 2020 HADES Seminar, UC Berkeley  
 Dec 2019 Graduate Student Seminar, Microlocal Analysis Program, MSRI

## Recent Conferences and Workshops Attended

Mar 2026 “AMS Spring Eastern Sectional Meeting”, Boston College  
 Feb 2026 “Modern Applications of Microlocal Analysis”, University College London, UK  
 Jan 2026 “High energy spectral theory: geometry and dynamics”, Isaac Newton Institute, UK  
 Jan 2026 “Joint Mathematics Meeting”, Washington, DC  
 Jun 2025 “Great Lakes Mathematical Physics Meeting”, University of Kentucky  
 Mar 2025 “AMS Spring Central Sectional Meeting”, University of Kansas  
 Dec 2024 “Harmonic and Microlocal Analysis in Partial Differential Equations”, MATRIX Institute, Australia  
 Oct 2024 “AMS Fall Western Sectional Meeting”, UC Riverside  
 Oct 2024 “Triangle Area Inverse Problems Weekend”, NC State University  
 Aug 2024 “Summer School: Geometric Inverse Problems and Inverse Problems for Elliptic Equations”, UC Santa Cruz  
 Jul 2024 “NU Trends in Ergodic Theory”, Northwestern University  
 Jun 2024 “Microlocal Analysis and Quantum Dynamics’, Northwestern University  
 Jun 2024 “Great Lakes Mathematical Physics Meeting”, Michigan State University  
 May 2024 “SIAM Conference on Imaging Sciences (IS24)”, Atlanta, GA  
 May 2024 “From Microlocal to Global Analysis @ MIT”, MIT  
 Mar 2024 “13th Ohio River Analysis Meeting”, University of Kentucky

## Service to Profession

Peer reviewer for the following journals:

- Inverse Problems (2 articles refereed)
- Transactions of the American Mathematical Society (1 article refereed)
- SIAM Journal on Mathematical Analysis (1 article refereed)
- Inverse Problems and Imaging (1 article refereed)
- Journal of Functional Analysis (1 article refereed)
- Rocky Mountain Journal of Mathematics (1 article refereed)

Ph.D. defense committees served on:

- Nicholas Lohr, Northwestern University, Apr 2025

## Mentoring and Outreach

Summer 2025 Northwestern Dynamics RTG REU, Northwestern University  
*–Mentored an undergraduate research project on quantum trajectories and quantum-classical correspondences. Served on a panel on “How to Give a Talk.”*

|                  |      |                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June             | 2025 | Career Development Panel at GLaMP 2025<br>– <i>Served on a panel aimed at early-career mathematicians discussing career development issues and questions at the workshop “Great Lakes Mathematical Physics Meeting 2025.”</i>                                                                                                                                                 |
| Summer           | 2024 | Microlocal Analysis and Quantum Dynamics Summer School<br>– <i>Organized summer school for undergraduates, graduate students, and early career researchers, covering topics in semiclassical analysis and applications, in preparation for the conference of the same name. Planned logistics for lectures, problem sessions, and social activities for the participants.</i> |
| Autumn           | 2021 | Directed Reading Program, University of California, Santa Cruz<br>– <i>Directed undergraduate reading project in Fourier analysis.</i>                                                                                                                                                                                                                                        |
| Autumn to Spring | 2019 | TA Mentoring Program, Stanford University (5 quarters)                                                                                                                                                                                                                                                                                                                        |
|                  | 2021 | – <i>Mentored first-time teaching assistants by observing sections and providing feedback.</i>                                                                                                                                                                                                                                                                                |
| Spring to Spring | 2017 | Directed Reading Program, Stanford University (9 quarters)                                                                                                                                                                                                                                                                                                                    |
|                  | 2021 | – <i>Directed undergraduate reading projects in Fourier analysis, complex analysis, ergodic theory, geometric measure theory, Ramsey theory, Markov chains, and distribution theory.</i>                                                                                                                                                                                      |
| September        | 2020 | Workshop on best teaching practices for graduate students, Stanford University<br>– <i>Moderated a panel regarding effective strategies for being an effective TA.</i>                                                                                                                                                                                                        |
| Summer           | 2016 | Summer Analysis Bootcamp, University of Chicago<br>– <i>Teaching assistant for summer program for advanced undergraduates in analysis.</i>                                                                                                                                                                                                                                    |
| Summer           | 2013 | Young Scholars Program, University of Chicago<br>– <i>Teaching assistant for summer math program for high school students.</i>                                                                                                                                                                                                                                                |
| Summer           | 2013 | SESAME Program, University of Chicago<br>– <i>Teaching assistant for certification program for middle school mathematics teachers.</i>                                                                                                                                                                                                                                        |
| Spring           | 2013 | Neighborhood Schools Program, University of Chicago<br>– <i>Tutor for after-school program at local elementary schools</i>                                                                                                                                                                                                                                                    |

## Other Information

Languages (natural): Mandarin Chinese (native), English (native), Cantonese (basic)  
 Languages (computer): Python (proficient), LaTeX (proficient)  
 Citizenship: United States of America